

Stochastic Non-Linear Programming:

When some of the parameters, involved in the objective function and constraints of an NLPP, vary about their mean values, a general optimization problem has to be formulated as a stochastic non-linear programming problem. We here assume that all the random variables are independent and follow normal distribution. A stochastic programming problem can be stated in standard form as:

Find X which minimizes $f(Y)$ _____(1)

Such that $\Pr. [g_j(Y) \geq 0] \geq p_j, j = 1, 2, \dots, m$ _____(2)

where $Y \rightarrow$ vector of N random variables $y_1, y_2, \dots, y_N$ and it includes the decision variables $x_1, x_2, \dots, x_n$. The case when X is deterministic can be obtained as special case of the present formulation. Equation (2) denotes that the problem of realizing $g_j(Y)$ greater than or equal to zero must be $\geq$ the specified probability p_j . The problem stated that in (1) and (2) can be converted into an equivalent deterministic non-linear programming problem by applying the chance constrained programming technique as follows:

Objective function:

Expanding $f(Y)$ by Taylor's series about the mean values $\bar{y}_i$ of y_i , we get

$$f(Y) = f(\bar{Y}) + \sum_{i=1}^N (y_i - \bar{y}_i) \left. \frac{\partial f}{\partial y_i} \right|_{\bar{Y}} + \text{higher order terms} \quad \text{_____}(3)$$

Assuming the standard deviation of y_i, σ_{y_i} to be small, above can be written as,

$$f(Y) \cong f(\bar{Y}) - \sum_{i=1}^N \bar{y}_i \left(\left. \frac{\partial f}{\partial y_i} \right|_{\bar{Y}} \right) + \sum_{i=1}^N y_i \left(\left. \frac{\partial f}{\partial y_i} \right|_{\bar{Y}} \right) = \psi(Y) \text{ (say)} \quad \text{_____}(4)$$

If all $y_i (i = 1, 2, \dots, N)$ follows normal distribution, $\psi(Y)$ which is a linear function of Y , also follows normal distribution. The mean and variance of ψ are given by,

$$\text{Mean } \bar{\psi} = \psi(\bar{Y}) \cong f(\bar{Y}) \quad \text{_____}(5)$$

$$\text{Var}(\psi) = \sigma_{\psi}^2 = E\{\psi(Y) - \psi(\bar{Y})\}^2$$

$$\begin{aligned} &= E \left\{ f(\bar{Y}) + \sum_{i=1}^N (y_i - \bar{y}_i) \left. \frac{\partial f}{\partial y_i} \right|_{\bar{Y}} - f(\bar{Y}) \right\}^2 \\ &= E \left\{ \sum_{i=1}^N (y_i - \bar{y}_i) \left. \frac{\partial f}{\partial y_i} \right|_{\bar{Y}} \right\}^2 \end{aligned}$$

$$= \sum_{i=1}^N \left(\frac{\partial f}{\partial y_i} \Big|_{\bar{Y}} \right)^2 E(y_i - \bar{y}_i)^2, \text{ [since } E\{(y_i - \bar{y}_i)(y_k - \bar{y}_k)\} = \text{Cov.}(y_i, y_k) \\ = 0, \text{ as all } y_i \text{ are independent}]$$

$$= \sum_{i=1}^N \left(\frac{\partial f}{\partial y_i} \Big|_{\bar{Y}} \right)^2 \sigma_{y_i}^2$$

Thus,
$$\text{Var}(\psi) = \sum_{i=1}^N \left(\frac{\partial f}{\partial y_i} \Big|_{\bar{Y}} \right)^2 \sigma_{y_i}^2 \text{ (6)}$$

The new deterministic objective function becomes,

$$F(Y) = K_1 \bar{\psi} + K_2 \sigma_{\psi} \text{ (7)}$$

where, $K_1 \geq 0$, $K_2 \geq 0$ and their numerical values indicate the relative importance of $\bar{\psi}$ and σ_{ψ} for minimization.

Constraints:

Here the constraints are $\text{Pr.}[g_j(Y) \geq 0] \geq p_j$, $j = 1, 2, \dots, m$. (by (2))

If some parameters are random in nature, the constraints will be probabilistic in nature. The constraint (2) can be written as,

$$\int_0^{\infty} f_{g_j}(g_j) dg_j \geq p_j \text{ (8), [since } \text{Pr.}(X \leq x) = \int_{-\infty}^x f(x) dx, \\ \text{by definition, where } f(x) \text{ is the density function of random variable } X]$$

where, $f_{g_j}(g_j)$ is the probability density function of random variable g_j (a function of several random variables is also a random variable) whose range is assumed to be $-\infty$ to ∞ . The constraint $g_j(Y)$ can be expanded about the vector $\bar{Y}$, mean values of random variable Y using Taylor series expansion as

$$g_j(Y) \cong g_j(\bar{Y}) + \sum_{i=1}^N (y_i - \bar{y}_i) \frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \text{ (9)}$$

Therefore, mean and standard deviation of g_j i. e., $\bar{g}_j$ and σ_{g_j} respectively can be obtained as,

$$\bar{g}_j = g_j(\bar{Y}) \text{ (10)}$$

Since,

$$\bar{g}_j(Y) = E\{g_j(Y)\}$$

$$\begin{aligned}
&= E \left\{ g_j(\bar{Y}) + \sum_{i=1}^N (y_i - \bar{y}_i) \frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \right\} \\
&= g_j(\bar{Y}) - \sum_{i=1}^N \bar{y}_i \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \right) + \sum_{i=1}^N E(y_i) \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \right) \\
&= g_j(\bar{Y}) - \sum_{i=1}^N \bar{y}_i \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \right) + \sum_{i=1}^N \bar{y}_i \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \right) \\
&= g_j(\bar{Y})
\end{aligned}$$

and,

$$\begin{aligned}
\sigma_{g_j}^2 &= E \{ g_j(Y) - g_j(\bar{Y}) \}^2 \\
&= E \left\{ g_j(\bar{Y}) + \sum_{i=1}^N (y_i - \bar{y}_i) \frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} - g_j(\bar{Y}) \right\}^2 \\
&= \sum_{i=1}^N E(y_i - \bar{y}_i)^2 \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \right)^2 \\
\sigma_{g_j} &= \sqrt{\sum_{i=1}^N E(y_i - \bar{y}_i)^2 \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{Y}} \right)^2} \quad \text{_____ (11) (Since, all covariances are zero)}
\end{aligned}$$

Let us now introduce a new variable

$$\theta = \frac{g_j(Y) - g_j(\bar{Y})}{\sigma_{g_j}} \quad \text{_____ (12)}$$

Eqn. (2) gives $Pr. [g_j(Y) \geq 0] \geq p_j, j = 1, 2, \dots, m$

or, $Pr. \left[\frac{g_j - \bar{g}_j}{\sigma_{g_j}} \geq -\frac{\bar{g}_j}{\sigma_{g_j}} \right] \geq p_j, j = 1, 2, \dots, m$ _____ (12a)

Since for normal distribution $Pr. (-\infty < t < \infty) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = 1$ _____ (13)

where t is a standard normal variate, therefore by (12a) and using (13), we have,

$$\int_{-\frac{\bar{g}_j}{\sigma_{g_j}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\theta^2/2} d\theta \geq p_j$$

$$\text{or,} \quad \int_{-\frac{\bar{g}_j}{\sigma_{g_j}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\theta^2/2} d\theta \geq \varphi_j(s_j), \quad [p_j = \varphi_j(s_j) \text{ (say)}]$$

$$= 1 - \varphi_j(-s_j)$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt - \int_{-\infty}^{-s_j} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

$$\text{or,} \quad \int_{-\frac{\bar{g}_j}{\sigma_{g_j}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\theta^2/2} d\theta \geq \int_{-s_j}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Let us denote $s_j = \chi_j(p_j)$

where, $\chi_j(p_j)$ is the value of the standard normal variate corresponding to the probability p_j .
Thus,

$$, \quad \int_{-\frac{\bar{g}_j}{\sigma_{g_j}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\theta^2/2} d\theta \geq \int_{-\chi_j(p_j)}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \text{ which holds only if}$$

$$-\frac{\bar{g}_j}{\sigma_{g_j}} \leq -\chi_j(p_j)$$

$$\text{or,} \quad -\bar{g}_j + \sigma_{g_j}(\chi_j(p_j)) \leq 0 \text{ _____(15)}$$

$$i.e., \quad \bar{g}_j - (\chi_j(p_j)) \left(\sum_{i=1}^N \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{y}} \right)^2 \sigma_{y_i}^2 \right)^{\frac{1}{2}} \geq 0, \quad j = 1, 2, \dots, m \text{ _____(16)}$$

Thus, optimization problem (1) and (2) can be stated in its equivalent deterministic form as,

$$\text{Minimize } F(Y) = K_1 \bar{\psi} + K_2 \sigma_{\psi}$$

such that

$$\bar{g}_j - \chi_j(p_j) \left(\sum_{i=1}^N \left(\frac{\partial g_j}{\partial y_i} \Big|_{\bar{y}} \right)^2 \sigma_{y_i}^2 \right)^{\frac{1}{2}} \geq 0, \quad j = 1, 2, \dots, m$$

Example 4:

A company manufactures two products P_1 and P_2 . The details of machine-hour requirements, profit per unit and the maximum available machine hours of the two machine types are summarized in the following table. Each non-deterministic data follows normal distribution and its mean and variance are given in the same table. Formulate a chance-constrained

programming model for this situation to find the production volume of each product such that the total profit is maximized.

Machine Type	Product		Maximum hours available	Probability
	P_1	P_2		
1	$a_{11}=8$ μ_{11} σ_{11}^2	a_{12} $\mu_{12} = 6$ $\sigma_{12}^2 = 25$	b_1 $\mu_1 = 96$ $\sigma_1^2 = 64$	0.95
2	a_{21} $\mu_{21} = 4$ $\sigma_{21}^2 = 4$	$a_{22} = 12$	100	0.90
Profit per unit	Rs. 50	Rs. 65		

Example 17.11 Develop a deterministic version of the following stochastic model:

subject to

$$\text{Maximize } Z = X_1 + 2X_2 + 5X_3$$

$$P(10X_1 + 15X_2 + 2X_3 \leq b_1) \geq 0.80 \quad (17.63)$$

$$P(4X_1 + a_2X_2 + a_3X_3 \leq 20) \geq 0.90 \quad (17.64)$$

$$X_1, X_2, X_3 \geq 0$$

In this model b_1 , a_2 and a_3 are random variables following normal distribution. The mean and variance of each of these random variables are summarized in Table 17.22.

Table 17.22 Mean and Variance of Random Variables

Random variable	Mean, μ	Variance, σ^2
b_1	40	25
$a_{22} \leftarrow a_2$	6	4
$a_{23} \leftarrow a_3$	3	2

STOCHASTIC GEOMETRIC PROGRAMMING

The deterministic geometric programming problem has been already considered. If the constant involved in the posynomials are random variables, the chance constrained programming methods, discussed earlier, can be applied to this problem. The probabilistic geometric programming problem can be stated as follows:

Find $X = \{x_1 \ x_2 \ \dots \dots x_n\}^T$ which minimizes $f(Y)$

Subject to

$$P[g_j(Y) > 0] \geq p_j, \quad j=1,2,\dots,m \quad \dots\dots\dots(1)$$

where $Y = \{y_1 \ y_2 \ \dots \dots y_N\}^T$ is the vector of N random variables (may include the variables $x_1, x_2, \dots x_n$) and $f(Y)$ and $g_j(Y)$, $j=1,2,\dots,m$, are posynomials. By expanding the objective function about the mean values of the random variables $y_i, \bar{y}_i$ and retaining only the first two terms, we can express the mean and variance of $f(Y)$ as done earlier. Thus the new objective function, $F(Y)$ can be expressed as

$$F(Y) = k_1 \bar{\psi} + k_2 \sigma_{\psi} \quad \dots\dots\dots(2)$$

The probabilistic constraints of eq. (1) can be converted into deterministic form as shown earlier:

$$\bar{g}_j - \phi_j(p_j) \left[\sum_{i=1}^N \left(\frac{\partial g_j}{\partial y_i} \right)^2 \sigma_{y_i}^2 \right]^{\frac{1}{2}} \geq 0, \quad j=1,2,\dots,m \quad (3)$$

Thus the optimization problem of eq (1) can be stated equivalently as:

Find Y which minimizes $F(Y)$ given by eq. (2) subject to the constraints of (3)

Example: A factory manufactures products A, B and C . Each of these products is processed through different production stages . The times required to manufacture 1 unit of each of the three products at different stages and the daily capacity of the stages are probabilistic with means and standard deviations as indicated below.

Time per Unit (min) for product

	<u>A</u>		<u>B</u>		<u>C</u>		<u>Stage Capacity (mins/day)</u>	
<u>Stage</u>	<u>Mean</u>	<u>Standard Deviation</u>	<u>Mean</u>	<u>Standard Deviation</u>	<u>Mean</u>	<u>S.D.</u>	<u>Mean</u>	<u>S.D.</u>
1	4	1	8	3	4	4	1720	172
2	12	2	0	0	8	2	1840	276
3	4	2	16	4	0	0	1680	336

The Profit per unit is also a random variable with the following data:

Profit (\$)

<u>Product</u>	<u>Mean</u>	<u>Standard Deviation</u>
A	6	2
B	4	1
C	10	3

Assuming that all amounts produced are absorbed by the market, **formulate** the problem to determine the daily number of units to be manufactured of each product for the following cases.

- (a) The object is to maximize the expected profit.
- (b) The objective is to maximize the standard deviation of the profit
- (c) The objective is to maximize the sum of expected profit and the standard deviation of the profit.

Assume that all the random variables follow normal distribution and the constraints have to be satisfied with a probability of atleast 0.95.